Physics 828: Homework Set No. 1

Due date: Friday, January 13, 2011, 1:00pm in PRB M2043 (Biao Huang's office)

Total point value of set: 60 points

Problem 1 (10 pts.): Exercise 9.4.1 (Shankar, p. 244)

Problem 2 (10 pts.): Exercise 9.4.2 (Shankar, p. 244)

Problem 3 (10 pts.): Exercise 9.4.3 (Shankar, p. 244)

Problem 4 (10 pts.): Exercise 9.4.4 (Shankar, p. 244)

Problem 5 (10 pts.): Exercise 10.1.1 (Shankar, p. 251)

Problem 6 (10 pts.): Exercise 10.1.2 (Shankar, p. 251)

Exercise 9.4.1 (p. 244)

$$\langle 1|\frac{1}{\hat{X}^2}|1\rangle = \int_{-\infty}^{\infty} dx \, \underbrace{A_1 H_1\left(\sqrt{\frac{m\omega}{\hbar}}\,x\right)}_{\left(\frac{m\omega}{\pi\hbar}\right)^{1/4}\sqrt{\frac{1}{2}}\,2\sqrt{\frac{m\omega}{\hbar}}\,x} e^{-\frac{m\omega}{2\hbar}x^2} \frac{1}{x^2} \underbrace{A_1 H_1\left(\sqrt{\frac{m\omega}{\hbar}}\,x\right)}_{\left(\frac{m\omega}{\pi\hbar}\right)^{1/4}\frac{1}{\sqrt{2}}\,2\sqrt{\frac{m\omega}{\hbar}}\,x} e^{-\frac{m\omega}{2\hbar}x^2}$$

$$= \sqrt{\frac{m\omega}{\pi\hbar}}\, 2\frac{m\omega}{\hbar}\int_{-\infty}^{\infty} dx \, e^{-\frac{m\omega}{2\hbar}x^2} = \frac{2}{\sqrt{\pi}}\frac{m\omega}{\hbar}\underbrace{\int_{-\infty}^{\infty} d\xi \, e^{-\xi^2}}_{\sqrt{\pi}} = 2\,\frac{m\omega}{\hbar}$$

$$\langle 1|\hat{X}^2|1\rangle = \frac{\hbar}{2m\omega}\langle 1|(\hat{a}+\hat{a}^+)^2|1\rangle = \frac{\hbar}{2m\omega}\langle 1|\underbrace{\hat{a}^2}_{0} + \underbrace{\hat{a}\hat{a}^+}_{1+\hat{a}^+\hat{a}} + \hat{a}^+\hat{a} + \underbrace{\hat{a}^{+2}}_{0}|1\rangle$$

$$= \frac{\hbar}{2m\omega}\langle 1|2\hat{n}+1|1\rangle \quad (\hat{n} = \hat{a}^+\hat{a})$$

$$= \frac{3}{2}\frac{\hbar}{m\omega} \implies \frac{1}{\langle 1|\hat{X}^2|1\rangle} = \frac{2}{3}\frac{m\omega}{\hbar}$$

$$\implies \boxed{\langle 1|\frac{1}{\hat{X}^2}|1\rangle \simeq \frac{1}{\langle 1|\hat{X}^2|1\rangle} \simeq \frac{m\omega}{\hbar}} \checkmark$$

The order-of-magnitude estimates are accurate within about a factor 3:

$$\boxed{\langle 1|\frac{1}{\hat{X}^2}|1\rangle = 3\,\frac{1}{\langle 1|\hat{X}^2|1\rangle} = 2\,\frac{m\omega}{\hbar}}$$

Exercise 9.4.2 (p.244)

(1) $\langle\psi|\psi\rangle = \int d^3r \langle\psi|\vec{r}\rangle\langle\vec{r}|\psi\rangle = \int d\Omega \int_0^\infty r^2 dr\, |\psi(\vec{r})|^2$

$$= 4\pi \int_0^\infty r^2 dr \frac{1}{\pi a_0^3} e^{-2r/a_0}$$

$$\overset{(p.660)}{=} \frac{4}{a_0^3} \int_0^\infty dr\, r^2 e^{-\frac{2}{a_0}\cdot r} = \frac{4}{a_0^3} \frac{2!}{(2/a_0)^3} = 1 \checkmark$$

(2) $(\Delta X)^2 = \langle\psi|\hat{X}^2|\psi\rangle - \langle\psi|\hat{X}|\psi\rangle^2$

Since $\psi(\vec{r})$ is spherically symmetric (it depends only on $r = \sqrt{x^2+y^2+z^2}$, it is reflection symmetric in x ($\psi(x) = \psi(-x)$), and hence $\langle\psi|\hat{X}|\psi\rangle = 0$.

Because $\psi(r)$ depends only on the combination $r = \sqrt{x^2+y^2+z^2}$, it is obvious that

$$\langle\psi|\hat{X}^2|\psi\rangle = \langle\psi|\hat{Y}^2|\psi\rangle = \langle\psi|\hat{Z}^2|\psi\rangle = \tfrac{1}{3}\langle\psi|\hat{X}^2+\hat{Y}^2+\hat{Z}^2|\psi\rangle$$

$$= \frac{1}{3}\int d^3r\, (x^2+y^2+z^2)\psi^2(r)$$

$$= \frac{1}{3} 4\pi \int_0^\infty r^2 dr\, r^2 \frac{1}{\pi a_0^3} e^{-2r/a_0}$$

$$= \frac{4}{3}\frac{1}{a_0^3}\int_0^\infty dr\, r^4 e^{-\frac{2}{a_0}r} \overset{p.660}{=} \frac{4}{3}\frac{1}{a_0^3}\cdot\frac{4!}{(2/a_0)^5}$$

$$= \underbrace{\frac{4\cdot 24}{3\cdot 32}}_{1} a_0^2 = a_0^2 \qquad \Longrightarrow (\Delta X)^2 = a_0^2$$

Hence

$$\Delta X = a_0 = \frac{\hbar^2}{me^2} \quad \text{as stated in Eq. (9.4.9)} \checkmark$$

(3) $\langle\psi|\frac{1}{r}|\psi\rangle = \int d^3r\, \frac{1}{r}\, \frac{1}{\pi a_0^3} e^{-\frac{2}{a_0}r}$

$$= \frac{4}{a_0^3}\int_0^\infty r^2 dr\, \frac{1}{r}\, e^{-\frac{2}{a_0}r}$$

$$= \frac{4}{a_0^3}\int_0^\infty dr\, r\, e^{-\frac{2}{a_0}r} \overset{(p.\,660)}{=} \frac{4}{a_0^3}\cdot\frac{1!}{(2/a_0)^2} = \frac{1}{a_0}$$

$$= \frac{me^2}{\hbar^2}$$

$$\langle\psi|r|\psi\rangle = \frac{4}{a_0^3}\int_0^\infty dr\, r^3 e^{-\frac{2}{a_0}r} = \frac{4}{a_0^3}\,\frac{3!}{(2/a_0)^4} = \frac{3}{2}a_0$$

$$\Rightarrow \frac{1}{\langle\psi|r|\psi\rangle} = \frac{2}{3a_0} = \frac{2}{3}\frac{me^2}{\hbar^2}$$

Hence $\langle\frac{1}{r}\rangle \simeq \frac{1}{\langle r\rangle} \simeq \frac{me^2}{\hbar^2}$ ✓ with the order-of-magnitude estimate being accurate within a factor 1.5:

$$\langle\frac{1}{r}\rangle = \frac{me^2}{\hbar^2} = \frac{3}{2}\frac{1}{\langle r\rangle}.$$

Exercise 9.4.3 (p. 244)

$$\hat{H} = \frac{\hat{P}^2}{2m} - \frac{e^2}{\sqrt{\hat{R}^2}} \quad \text{in 1 dimension } R.$$

Assume $\Delta P \cdot \Delta R \geq \frac{\hbar}{2}$

$$\langle \hat{H} \rangle = \frac{\overbrace{\langle \hat{P}^2 \rangle}^{=(\Delta P)^2 + \langle \hat{P} \rangle^2}}{2m} - \left\langle \frac{e^2}{\sqrt{\hat{R}^2}} \right\rangle \geq \frac{(\Delta P)^2}{2m} - \left\langle \frac{e^2}{\sqrt{\hat{R}^2}} \right\rangle$$

$$\gtrsim \frac{(\Delta P)^2}{2m} - \frac{e^2}{\langle \sqrt{\hat{R}^2} \rangle} \simeq \frac{(\Delta P)^2}{2m} - \frac{e^2}{\sqrt{\underbrace{\langle \hat{R}^2 \rangle}_{=(\Delta R)^2 + \langle \hat{R} \rangle^2}}}$$

$$\geq \frac{(\Delta P)^2}{2m} - \frac{e^2}{\Delta R}$$

$$\geq \frac{\hbar^2/4}{2m} \frac{1}{(\Delta R)^2} - \frac{e^2}{\Delta R} = \frac{\hbar^2}{8m} \left(\frac{1}{(\Delta R)^2} - \frac{8me^2}{\hbar^2 \Delta R} \right)$$

The right hand side is minimal when

$$-2 \frac{1}{(\Delta R)^3} + \frac{8me^2}{\hbar^2} \frac{1}{(\Delta R)^2} = 0 \Rightarrow \frac{4me^2}{\hbar^2} \Delta R = 1$$

i.e. for $\Delta R = \frac{\hbar^2}{4me^2}$

Hence

$$\langle \hat{H} \rangle \gtrsim \frac{\hbar^2}{8m} \left(\frac{16 m^2 e^4}{\hbar^4} - \frac{8me^2}{\hbar^2} \frac{4me^2}{\hbar^2} \right)$$

$$= -\frac{16\hbar^2}{8m} \frac{m^2 e^4}{\hbar^4} = -2 \frac{me^4}{\hbar^2}$$

The estimated ground state energy is

$E_g \simeq -2 \frac{me^4}{\hbar^2}$, which is 9 times larger in magnitude than the 3-d estimate (9.4.7)

Exercise 9.4.4 (p. 244)

$$\hat{T} = \frac{\hat{P}^2}{2m} \qquad \left(\hat{P} = \hat{P}_x\ ,\ [\hat{P},\hat{X}] = -i\hbar\hat{\mathbb{1}}\right)$$

$$(\Delta T)^2(\Delta X)^2 \geq \frac{1}{4}\left[\langle\psi|[\hat{T}-\frac{\overline{p^2}}{2m},\hat{X}-\bar{x}]_+|\psi\rangle^2 + \langle\psi|\frac{1}{i}[\frac{\hat{P}^2}{2m},\hat{X}]_-|\psi\rangle^2\right]$$

$$(\overline{p^2} \equiv \langle\hat{P}^2\rangle\ ,\ \bar{x} = \langle\hat{x}\rangle)$$

- $$\left[\frac{\hat{P}^2}{2m},\hat{X}\right]_- = \frac{1}{2m}[\hat{P}^2,\hat{X}] = \frac{1}{2m}\left([\hat{P},\hat{X}]\hat{P} + \hat{P}\underbrace{[\hat{P},\hat{X}]}_{-i\hbar}\right) = -\frac{i\hbar}{m}\hat{P}$$

$$\Rightarrow \langle\psi|\frac{1}{i}[\frac{\hat{P}^2}{2m},\hat{X}]|\psi\rangle^2 = \frac{\hbar^2}{m^2}\langle\hat{P}\rangle^2 = \frac{\hbar^2}{m^2}\bar{p}^2 \qquad (\bar{p} = \langle\hat{P}\rangle)$$

- $$\left[\hat{T}-\frac{\overline{p^2}}{2m},\hat{X}-\bar{x}\right]_+ = \frac{\hat{P}^2\hat{X}+\hat{X}\hat{P}^2}{2m} - \frac{\overline{p^2}}{2m}\hat{X} - \frac{\bar{x}}{m}\hat{P}^2 + \bar{x}\frac{\overline{p^2}}{2m}$$

$$\Rightarrow \langle\psi|[\hat{T}-\frac{\overline{p^2}}{2m},\hat{X}-\bar{x}]|\psi\rangle = \frac{1}{m}\left[\langle\frac{\hat{P}^2\hat{X}+\hat{X}\hat{P}^2}{2}\rangle - \bar{x}\,\overline{p^2}\right]$$

$$\hat{P}^2\hat{X}+\hat{X}\hat{P}^2 = \hat{P}\underbrace{[\hat{P},\hat{X}]}_{-i\hbar} + \hat{P}\hat{X}\hat{P} + \underbrace{[\hat{X},\hat{P}]}_{i\hbar}\hat{P} + \hat{P}\hat{X}\hat{P} = 2\hat{P}\hat{X}\hat{P}$$

$$\Rightarrow \boxed{(\Delta T)^2(\Delta X)^2 \geq \frac{1}{4m^2}\left[\left(\langle\hat{P}\hat{X}\hat{P}\rangle - \bar{x}\,\overline{p^2}\right)^2 + \hbar^2\bar{p}^2\right]} \quad (\bar{p} \equiv \langle\hat{p}\rangle)$$

$$= \frac{1}{4m^2}\left[\langle\hat{P}\hat{X}\hat{P}\rangle^2 - 2\bar{x}\,\overline{p^2}\langle\hat{P}\hat{X}\hat{P}\rangle + \bar{x}^2(\overline{p^2})^2 + \hbar^2\bar{p}^2\right]$$

For any reflection symmetric state ($\psi(x) = \psi(-x)$, $\psi(p) = \psi(-p)$, $\psi(x)$ real) we have $\bar{p} = 0 = \bar{x}$. For such states the relation reads

$$\Delta T \cdot \Delta X \geq |\langle\psi|\hat{P}\hat{X}\hat{P}|\psi\rangle| = \left|\int_{-\infty}^{\infty} dp\ p\,\psi(p)\,i\hbar\frac{d}{dp}(p\psi(p))\right| =$$

$$= |\langle \hat{P}\psi | \hat{X} | \hat{P}\psi \rangle| = \left| \int_{-\infty}^{\infty} dx\, x\, \hbar^2 \left(\frac{\partial \psi}{\partial x}\right)^2 \right|$$

For any even or odd function $\left|\frac{\partial \psi}{\partial x}\right|^2$ is even;

then $\int_{-\infty}^{\infty} dx\, x \left(\frac{\partial \psi}{\partial x}\right)^2 = 0$. So we can find states for which the r.h.s. of the uncertainty product inequality vanishes, and we are left with the trivial statement

$$\boxed{\Delta T \cdot \Delta X \geq 0}$$

(trivial, since the l.h.s. is always positive by definition).

We conclude that $\Delta T \cdot \Delta X$ has no non-trivial universal lower bound. This makes this uncertainty relation pretty useless.

Exercise 10.1.1 (p. 251)

(1) $\left[\hat{\Omega}_1 \otimes \hat{\mathbb{1}}_2, \mathbb{1}_1 \otimes \hat{\Lambda}_2\right] = \left(\hat{\Omega}_1 \otimes \hat{\mathbb{1}}_2\right)\left(\mathbb{1}_1 \otimes \hat{\Lambda}_2\right) - \left(\mathbb{1}_1 \otimes \hat{\Lambda}_2\right)\left(\hat{\Omega}_1 \otimes \hat{\mathbb{1}}_2\right)$

$$\overset{(2)}{=} \left(\hat{\Omega}_1 \mathbb{1}_1 \otimes \mathbb{1}_2 \hat{\Lambda}_2\right) - \left(\mathbb{1}_1 \hat{\Omega}_1 \otimes \hat{\Lambda}_2 \hat{\mathbb{1}}_2\right)$$

$$= \hat{\Omega}_1 \otimes \hat{\Lambda}_2 - \hat{\Omega}_1 \otimes \hat{\Lambda}_2 = 0 \quad \checkmark$$

(2) $\left(\hat{\Omega}_1 \otimes \hat{\Gamma}_2\right)\left(\hat{\theta}_1 \otimes \hat{\Lambda}_2\right)\left(|\psi_1\rangle \otimes |\psi_2\rangle\right) \overset{(10.1.14)}{=} \left(\hat{\Omega}_1 \otimes \hat{\Gamma}_2\right)\left(\left(\hat{\theta}_1 |\psi_1\rangle\right) \otimes \left(\hat{\Lambda}_2 |\psi_2\rangle\right)\right)$

$$\overset{(10.1.14)}{=} \left(\hat{\Omega}_1 \hat{\theta}_1 |\psi_1\rangle\right) \otimes \left(\hat{\Gamma}_2 \hat{\Lambda}_2 |\psi_2\rangle\right) \overset{(10.1.14)}{=} \left(\hat{\Omega}_1 \hat{\theta}_1 \otimes \hat{\Gamma}_2 \hat{\Lambda}_2\right)\left(|\psi_1\rangle \otimes |\psi_2\rangle\right)$$

This holds for any $|\psi_1\rangle \otimes |\psi_2\rangle$ (i.e. also for all basisstates in $\mathbb{V}_1 \otimes \mathbb{V}_2$), hence it is an operator identity:

$$\left(\hat{\Omega}_1 \otimes \hat{\Gamma}_2\right)\left(\hat{\theta}_1 \otimes \hat{\Lambda}_2\right) = \left(\hat{\Omega}_1 \hat{\theta}_1\right) \otimes \left(\hat{\Gamma}_2 \hat{\Lambda}_2\right)$$

(3) If $\left[\hat{\Omega}_1, \hat{\Lambda}_1\right] = \hat{\Gamma}_1$ then

$$\left[\hat{\Omega}_1^{1\otimes 2}, \hat{\Lambda}_1^{1\otimes 2}\right] = \left[\hat{\Omega}_1 \otimes \hat{\mathbb{1}}_2, \hat{\Lambda}_1 \otimes \hat{\mathbb{1}}_2\right] =$$

$$= \left(\hat{\Omega}_1 \otimes \hat{\mathbb{1}}_2\right)\left(\hat{\Lambda}_1 \otimes \hat{\mathbb{1}}_2\right) - \left(\hat{\Lambda}_1 \otimes \hat{\mathbb{1}}_2\right)\left(\hat{\Omega}_1 \otimes \hat{\mathbb{1}}_2\right)$$

$$= \hat{\Omega}_1 \hat{\Lambda}_1 \otimes \underbrace{\hat{\mathbb{1}}_2 \cdot \hat{\mathbb{1}}_2}_{\hat{\mathbb{1}}_2} - \hat{\Lambda}_1 \hat{\Omega}_1 \otimes \hat{\mathbb{1}}_2 \cdot \hat{\mathbb{1}}_2 = \left[\hat{\Omega}_1, \hat{\Lambda}_1\right] \otimes \hat{\mathbb{1}}_2 = \hat{\Gamma}_1 \otimes \hat{\mathbb{1}}_2 \quad \checkmark$$

Similarly $\left[\hat{\Omega}_2^{1\otimes 2}, \hat{\Lambda}_2^{1\otimes 2}\right] = \left[\hat{\mathbb{1}}_1 \times \hat{\Omega}_1, \hat{\mathbb{1}}_1 \otimes \hat{\Lambda}_2\right] = \dots = \mathbb{1}_1 \otimes \left[\hat{\Omega}_2, \hat{\Lambda}_2\right]$

$$= \mathbb{1}_1 \otimes \hat{\Gamma}_2$$

(4) $\left(\hat{\Omega}_1^{1\otimes 2} + \hat{\Omega}_2^{1\otimes 2}\right)^2 = \left(\hat{\Omega}_1 \times \hat{\mathbb{1}}_2 + \hat{\mathbb{1}}_1 \otimes \hat{\Omega}_2\right)^2$

$= \left(\hat{\Omega}_1 \times \hat{\mathbb{1}}_2\right)^2 + \left(\hat{\Omega}_1 \otimes \mathbb{1}_2\right)\left(\mathbb{1}_1 \otimes \hat{\Omega}_2\right) + \left(\hat{\mathbb{1}}_1 \otimes \hat{\Omega}_2\right)\left(\hat{\Omega}_1 \otimes \hat{\mathbb{1}}_2\right) + \left(\mathbb{1}_1 \otimes \hat{\Omega}_2\right)^2$

$= \hat{\Omega}_1^2 \otimes \hat{\mathbb{1}}_2 + 2\,\hat{\Omega}_1 \otimes \hat{\Omega}_2 + \hat{\mathbb{1}}_1 \otimes \hat{\Omega}_2^2$ ✓

Exercise 10.1.2 (p. 251)

$$\hat{\sigma}_1 \leftrightarrow \begin{pmatrix} a & b \\ c & d \end{pmatrix} = \begin{pmatrix} \sigma_1^{++} & \sigma_1^{+-} \\ \sigma_1^{-+} & \sigma_1^{--} \end{pmatrix} \text{ ; basis vectors } \left.\begin{matrix} |+\rangle_1 \leftrightarrow \begin{pmatrix} 1 \\ 0 \end{pmatrix} \\ |-\rangle_1 \leftrightarrow \begin{pmatrix} 0 \\ 1 \end{pmatrix} \end{matrix}\right\} \text{in } \mathbb{V}_1$$

$$\hat{\sigma}_2 \leftrightarrow \begin{pmatrix} e & f \\ g & h \end{pmatrix} \qquad \left.\begin{matrix} |+\rangle_2 \leftrightarrow \begin{pmatrix} 1 \\ 0 \end{pmatrix} \\ |-\rangle_2 \leftrightarrow \begin{pmatrix} 0 \\ 1 \end{pmatrix} \end{matrix}\right\} \text{in } \mathbb{V}_2$$

(1) $\hat{\sigma}_1^{1\otimes 2} = \hat{\sigma}_1 \otimes \hat{\mathbb{1}}_2 \longleftrightarrow$

$$\longleftrightarrow \begin{pmatrix} \langle ++|\hat{\sigma}_1\otimes\hat{\mathbb{1}}_2|++\rangle & \langle ++|\hat{\sigma}_1\otimes\mathbb{1}_2|+-\rangle & \langle ++|\hat{\sigma}_1\otimes\hat{\mathbb{1}}_2|-+\rangle & \langle ++|\hat{\sigma}_1\otimes\hat{\mathbb{1}}_2|--\rangle \\ \langle +-|\hat{\sigma}_1\otimes\hat{\mathbb{1}}_2|++\rangle & \dots & \dots & \langle +-|\hat{\sigma}_1\otimes\hat{\mathbb{1}}_2|--\rangle \\ \langle -+|\hat{\sigma}_1\otimes\hat{\mathbb{1}}_2|++\rangle & \dots & \dots & \langle -+|\hat{\sigma}_1\otimes\hat{\mathbb{1}}_2|--\rangle \\ \langle --|\hat{\sigma}_1\times\hat{\mathbb{1}}_2|++\rangle & \dots & \dots & \langle --|\hat{\sigma}_1\times\hat{\mathbb{1}}_2|--\rangle \end{pmatrix}$$

$$= \begin{pmatrix} \underbrace{\langle +|\hat{\sigma}_1|+\rangle_1}_{a}\,\underbrace{{}_2\langle +|\mathbb{1}_2|+\rangle_2}_{1} & \underbrace{\langle +|\hat{\sigma}_1|+\rangle}_{a}\underbrace{\langle +|\mathbb{1}_2|-\rangle}_{0} & \underbrace{\langle +|\hat{\sigma}_1|-\rangle}_{b}\underbrace{\langle +|\mathbb{1}_2|+\rangle}_{1} & \underbrace{\langle +|\hat{\sigma}_1|-\rangle}_{b}\underbrace{\langle +|\mathbb{1}_2|-\rangle}_{0} \\ \underbrace{\langle +|\hat{\sigma}_1|+\rangle}_{a}\,\underbrace{\langle -|\mathbb{1}_2|+\rangle}_{0} & \overset{\dots}{a\cdot 1} & \overset{\dots}{b\cdot 0} & \overset{\dots}{b\cdot 1} \\ \underbrace{\langle -|\hat{\sigma}_1|+\rangle}_{c}\,\underbrace{\langle +|\mathbb{1}|+\rangle}_{1} & \overset{\dots}{c\cdot 0} & \overset{\dots}{d\cdot 1} & \overset{\dots}{d\cdot 0} \\ \underbrace{\langle -|\hat{\sigma}_1|+\rangle}_{c}\,\underbrace{\langle -|\mathbb{1}_2|+\rangle}_{0} & \overset{\dots}{c\cdot 1} & \overset{\dots}{d\cdot 0} & \underset{d\cdot 1}{\langle -|\hat{\sigma}_1|-\rangle\langle -|\mathbb{1}_2|-\rangle} \end{pmatrix}$$

$$= \begin{pmatrix} a & 0 & b & 0 \\ 0 & a & 0 & b \\ c & 0 & d & 0 \\ 0 & c & 0 & d \end{pmatrix} = \begin{pmatrix} a\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} & b\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \\ c\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} & d\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \end{pmatrix}$$

(2) $\hat{\sigma}_2^{1\otimes 2} = \mathbb{1}_1 \otimes \hat{\sigma}_2 \longleftrightarrow$

$$\longleftrightarrow \begin{pmatrix} \langle ++|\hat{\mathbb{1}}\otimes\hat{\sigma}_2|++\rangle & \langle ++|\hat{\mathbb{1}}_1\times\hat{\sigma}_2|+-\rangle & \langle ++|\hat{\mathbb{1}}_1\otimes\hat{\sigma}_2|-+\rangle & \langle ++|\hat{\mathbb{1}}\otimes\hat{\sigma}_2|--\rangle \\ \langle +-|\hat{\mathbb{1}}_1\times\hat{\sigma}_2|++\rangle & \langle +-|\hat{\mathbb{1}}\times\hat{\sigma}_2|+-\rangle & --- & \langle +-|\hat{\mathbb{1}}\otimes\hat{\sigma}_2|--\rangle \\ \langle -+|\hat{\mathbb{1}}_1\otimes\hat{\sigma}_2|++\rangle & -- & -- & \langle -+|\hat{\mathbb{1}}\otimes\hat{\sigma}_2|--\rangle \\ \langle --|\hat{\mathbb{1}}_1\otimes\hat{\sigma}_2|++\rangle & --- & --- & \langle --|\hat{\mathbb{1}}_1\otimes\hat{\sigma}_2|--\rangle \end{pmatrix}$$

$$= \begin{pmatrix} \underbrace{{}_1\langle +|\hat{\mathbb{1}}_1|+\rangle_1}_{1}\underbrace{{}_2\langle +|\hat{\sigma}_2|+\rangle_2}_{e} & \underbrace{\langle +|\mathbb{1}_1|+\rangle}_{1}\underbrace{\langle +|\sigma_2|-\rangle}_{f} & \underbrace{\langle +|\mathbb{1}_1|-\rangle}_{0}\underbrace{\langle +|\hat{\sigma}_2|+\rangle}_{e} & \underbrace{\langle +|\hat{\mathbb{1}}_1|-\rangle}_{0}\underbrace{\langle +|\hat{\sigma}_2|-\rangle}_{f} \\ \underbrace{\langle +|\mathbb{1}_1|+\rangle}_{1}\underbrace{\langle -|\hat{\sigma}_2|+\rangle}_{g} & \underbrace{\langle +|\mathbb{1}|+\rangle}_{1}\underbrace{\langle -|\hat{\sigma}_2|-\rangle}_{h} & \overset{---}{0}\cdot g & \overset{--}{0}\cdot h \\ \underbrace{\langle -|\mathbb{1}_1|+\rangle}_{0}\underbrace{\langle +|\hat{\sigma}_2|+\rangle}_{e} & \langle -|\mathbb{1}_1|+\rangle\langle +|\hat{\sigma}_2|-\rangle \; 0\cdot f & \langle \cdots \; 1\cdot e & \underbrace{\langle -|\mathbb{1}_1|-\rangle}_{1}\underbrace{\langle +|\hat{\sigma}_2|-\rangle}_{f} \\ \underbrace{\langle -|\mathbb{1}_1|+\rangle}_{0}\underbrace{\langle -|\hat{\sigma}_2|+\rangle}_{g} & \overset{--}{0}\cdot h & \overset{--}{1}\cdot g & \underbrace{\langle -|\mathbb{1}_1|-\rangle}_{1}\underbrace{\langle -|\hat{\sigma}_2|-\rangle}_{h} \end{pmatrix}$$

$$= \begin{pmatrix} e & f & 0 & 0 \\ g & h & 0 & 0 \\ 0 & 0 & e & f \\ 0 & 0 & g & h \end{pmatrix} = \begin{pmatrix} \sigma_2 & 0 \\ 0 & \sigma_2 \end{pmatrix}$$

$$(3)\quad \hat{\sigma}_1 \otimes \hat{\sigma}_2 \longleftrightarrow \begin{pmatrix} \langle ++|\hat{\sigma}_1\otimes\hat{\sigma}_2|++\rangle & \langle ++|\hat{\sigma}_1\otimes\hat{\sigma}_2|+-\rangle & \langle ++|\hat{\sigma}_1\otimes\hat{\sigma}_2|-+\rangle & \langle ++|\hat{\sigma}_1\otimes\hat{\sigma}_2|--\rangle \\ \langle +-|\hat{\sigma}_1\otimes\hat{\sigma}_2|++\rangle & \langle +-|\hat{\sigma}_1\otimes\hat{\sigma}_2|+-\rangle & \cdots & \langle +-|\hat{\sigma}_1\otimes\hat{\sigma}_2|--\rangle \\ \langle -+|\hat{\sigma}_1\otimes\hat{\sigma}_2|++\rangle & \cdots & \cdots & \langle -+|\hat{\sigma}_1\otimes\hat{\sigma}_2|--\rangle \\ \langle --|\hat{\sigma}_1\otimes\hat{\sigma}_2|++\rangle & \cdots & \cdots & \langle --|\hat{\sigma}_1\otimes\hat{\sigma}_2|--\rangle \end{pmatrix}$$

$$= \begin{pmatrix} {}_1\langle +|\sigma_1|+\rangle_1\, {}_2\langle +|\hat{\sigma}_2|+\rangle_2, & \langle +|\hat{\sigma}_1|+\rangle\langle +|\hat{\sigma}_2|-\rangle, & \langle +|\hat{\sigma}_1|-\rangle\langle +|\hat{\sigma}_2|+\rangle, & \langle +|\sigma_1|-\rangle\langle +|\hat{\sigma}_2|-\rangle \\ \langle +|\hat{\sigma}_1|+\rangle\langle -|\hat{\sigma}_2|+\rangle, & \langle +|\hat{\sigma}_1|+\rangle\langle -|\hat{\sigma}_2|-\rangle, & \langle +|\hat{\sigma}_1|-\rangle\langle -|\hat{\sigma}_2|+\rangle, & \langle +|\sigma_1|-\rangle\langle -|\hat{\sigma}_2|-\rangle \\ \langle -|\hat{\sigma}_1|+\rangle\langle +|\hat{\sigma}_2|+\rangle, & \langle -|\hat{\sigma}_1|+\rangle\langle +|\sigma_2|-\rangle, & \langle -|\hat{\sigma}_1|-\rangle\langle +|\hat{\sigma}_2|+\rangle, & \langle -|\hat{\sigma}_1|-\rangle\langle +|\hat{\sigma}_2|-\rangle \\ \langle -|\sigma_1|+\rangle\langle -|\hat{\sigma}_2|+\rangle, & \langle -|\hat{\sigma}_1|+\rangle\langle -|\hat{\sigma}_2|-\rangle, & \langle -|\hat{\sigma}_1|-\rangle\langle -|\hat{\sigma}_2|+\rangle, & \langle -|\hat{\sigma}_1|-\rangle\langle -|\hat{\sigma}_2|-\rangle \end{pmatrix}$$

$$= \begin{pmatrix} ae & af & be & bf \\ ag & ah & bg & bh \\ ce & cf & de & df \\ cg & ch & dg & dh \end{pmatrix} \checkmark$$

Since $\hat{\sigma}_1 \otimes \hat{\sigma}_2 = \left(\hat{\sigma}_1 \hat{\sigma}_2\right)^{1\otimes 2}$, we can also get this result by multiplying the matrices from (1) and (2):

$$\left(\hat{\sigma}_1 \cdot \hat{\sigma}_2\right)^{1\otimes 2} \longleftrightarrow \begin{pmatrix} a & 0 & b & 0 \\ 0 & a & 0 & b \\ c & 0 & d & 0 \\ 0 & c & 0 & d \end{pmatrix}\begin{pmatrix} e & f & 0 & 0 \\ g & h & 0 & 0 \\ 0 & 0 & e & f \\ 0 & 0 & g & h \end{pmatrix} = \begin{pmatrix} ae & af & be & bf \\ ag & ah & bg & bh \\ ce & cf & de & df \\ cg & ch & dg & dh \end{pmatrix}$$

$$= \begin{pmatrix} e & f & 0 & 0 \\ g & h & 0 & 0 \\ 0 & 0 & e & f \\ 0 & 0 & g & h \end{pmatrix}\begin{pmatrix} a & 0 & b & 0 \\ 0 & a & 0 & b \\ c & 0 & d & 0 \\ 0 & c & 0 & d \end{pmatrix} \qquad \text{since } \left[\hat{\sigma}_1, \hat{\sigma}_2\right] = 0$$